NIOS Class 10 (211) Mathematics — Chapter-by-Chapter Fast Revision

NIOS Secondary (Class 10) Mathematics (211) — Exam Prep Study Kit

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> **Only the 13 Public Examination Chapters are included here.**
> Everything else in your textbook belongs to TMA and is excluded from the exam. Review these chapters one by one according to the 7-day study plan.

MODULE 1: ALGEBRA (20 Marks Total)

Chapter 4: Special Products and Factorisation

- **Key Concepts**:
- Identities: $(a \pm b)^2 = a^2 \pm 2ab + b^2$
- $(a + b)(a - b) = a^2 - b^2$
- $(a \pm b)^3 = a^3 \pm 3a^2b + 3ab^2 \pm b^3 = a^3 \pm b^3 \pm 3ab(a \pm b)$
- $a^3 + b^3 = (a + b)(a^2 - ab + b^2)$
- $a^3 - b^3 = (a - b)(a^2 + ab + b^2)$
- **Standard Solved Questions**:
- *Simplify* $(3x + 4y)^2 + (3x - 4y)^2$:
  - Use identity: $(a + b)^2 + (a - b)^2 = 2(a^2 + b^2)$.
  - Here $a = 3x, b = 4y \implies 2((3x)^2 + (4y)^2) = 2(9x^2 + 16y^2) = \mathbf{18x^2 + 32y^2}$.
- *Factorise* $a^6 - 64b^6$:
  - Write as difference of squares: $(a^3)^2 - (8b^3)^2 = (a^3 - 8b^3)(a^3 + 8b^3)$.
  - Now apply $x^3 - y^3$ and $x^3 + y^3$:
  - $a^3 - (2b)^3 = (a - 2b)(a^2 + 2ab + 4b^2)$
  - $a^3 + (2b)^3 = (a + 2b)(a^2 - 2ab + 4b^2)$

- **Final Answer**: $\mathbf{(a-2b)(a+2b)(a^2+2ab+4b^2)(a^2-2ab+4b^2)}$.

Chapter 5: Linear Equations

- **Key Concepts**:
- Elimination / Substitution methods to find (x, y).
- Graphical representation (see Template 2 in `04_GUARANTEED_SURE_SHOT_LONG_QUESTIONS.md`).
- **Standard Solved Question**:
- *Solve for x and y: $3x+2y=11$ and $2x+3y=4$:*
  - Multiply (1) by 3: $9x+6y=33$
  - Multiply (2) by 2: $4x+6y=8$
  - Subtract: $5x=25 \implies \mathbf{x=5}$.
  - Substitute $x=5$ into (1): $3(5)+2y=11 \implies 15+2y=11 \implies 2y=-4 \implies \mathbf{y=-2}$.

Chapter 6: Quadratic Equations

- **Key Concepts**:
- $ax^2+bx+c=0$
- Discriminant: $D=b^2-4ac$
- Roots: $x=\frac{-b\pm\sqrt{D}}{2a}$
- **Standard Solved Questions**:
- *Find the discriminant and nature of roots for $2x^2-6x+3=0$:*
  - $a=2, b=-6, c=3$
  - $D=(-6)^2-4(2)(3)=36-24=\mathbf{12}$.
  - Since $D>0$, the roots are **real and distinct**.
- *Find the roots of $x^2-5x+6=0$:*
  - $(x-2)(x-3)=0 \implies \mathbf{x=2,\ 3}$.

Chapter 7: Arithmetic Progression (AP)

- **Key Concepts**:
- General term: $a_n=a+(n-1)d$
- Sum of n terms: $S_n=\frac{n}{2}[2a+(n-1)d]$ or $S_n=\frac{n}{2}[a+l]$
- **Standard Solved Questions**:
- *Which term of the AP $3, 8, 13, 18\ldots$ is 78?:*

- $a = 3, d = 5, a_n = 78$
- $78 = 3 + (n-1)5 \implies 75 = 5(n-1) \implies n - 1 = 15 \implies \mathbf{n = 16}$.

- *Find the sum of the first 20 terms of the AP* $5, 9, 13, 17 \ldots$:
  - $a = 5, d = 4, n = 20$
  - $S_{20} = \frac{20}{2}[2(5) + (20-1) \times 4] = 10[10 + 76] = 10 \times 86 = \mathbf{860}$.

MODULE 2: COMMERCIAL MATHEMATICS (8 Marks)

Chapter 9: Installment Buying

- **Key Concepts**:
- $\text{Cash Price} = \text{Down Payment} + \text{Present Value of Installments}$
- $\text{Unpaid Balance} = \text{Cash Price} - \text{Cash Down Payment}$
- **Standard Solved Question**:
- *A table is sold for ₹1,700 cash or for ₹500 as cash down payment, followed by ₹1,400 after 2 months. Find the rate of interest charged*:
  - Cash Price $= 1700$
  - Cash Down Payment $= 500$
  - Principal for the loan (Unpaid balance) $= 1700 - 500 = ₹1,200$
  - Amount paid after 2 months $= ₹1,400$
  - Interest charged for 2 months $= 1400 - 1200 = ₹200$
  - Time $T = \frac{2}{12} = \frac{1}{6}$ year
  - Using $I = \frac{P \times R \times T}{100}$:

$$200 = \frac{1200 \times R \times \frac{1}{6}}{100} = 2R \implies R = \frac{200}{2} = \mathbf{100\%} \text{ per annum.}$$

MODULE 3: GEOMETRY (25 Marks Total)

Chapter 16 & 17: Circles, Angles, Secants & Tangents

- **Key Theorems**:
- Tangent is perpendicular to radius at point of contact ($\angle OPT = 90^\circ$).
- Tangents from an external point to a circle are equal in length ($PA = PB$).
- Angle subtended by an arc at the centre is double the angle at the circumference ($\angle AOC = 2\angle ABC$).

- Angle in a semicircle is 90°.
- Opposite angles of cyclic quadrilateral sum to 180°.
- **Standard Solved Question**:
- *In the given figure, O is the centre. PA and PB are tangents to the circle from P. If $\angle APB = 60^\circ$, find $\angle AOB$:*
  - In quadrilateral $OAPB$, $\angle OAP = 90^\circ$ and $\angle OBP = 90^\circ$.
  - $\angle AOB + \angle APB = 180^\circ \implies \angle AOB = 180^\circ - 60^\circ = \mathbf{120}^\circ$.

Chapter 18: Constructions

- **Core Problem**:
- Draw a circle of radius 3 cm or 4 cm. From point P at distance d from centre, draw tangents (refer to `04_GUARANTEED_SURE_SHOT_LONG_QUESTIONS.md` Template 3).

Chapter 19: Coordinate Geometry

- **Key Formulas**:
- Distance: $d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$
- Midpoint: $\left(\frac{x_1+x_2}{2}, \frac{y_1+y_2}{2}\right)$
- Section: $\left(\frac{mx_2+nx_1}{m+n}, \frac{my_2+ny_1}{m+n}\right)$
- **Standard Solved Question**:
- *Find the distance between $A(2, 3)$ and $B(6, 6)$:*
  - $d = \sqrt{(6-2)^2 + (6-3)^2} = \sqrt{4^2 + 3^2} = \sqrt{16+9} = \sqrt{25} = \mathbf{5}$ units.

MODULE 4: MENSURATION (10 Marks)

Chapter 21: Surface Area and Volume

- **Key Solids**:
- Cylinder: $V = \pi r^2 h,\ \text{CSA} = 2\pi rh$
- Cone: $V = \frac{1}{3}\pi r^2 h,\ l = \sqrt{r^2 + h^2},\ \text{CSA} = \pi rl$
- Sphere: $V = \frac{4}{3}\pi r^3,\ \text{Area} = 4\pi r^2$
- **Standard Solved Question**:
- *Find the volume and total surface area of a solid cylinder of radius 7 cm and height 10 cm (Take $\pi = \frac{22}{7}$):*

- $V = \pi r^2 h = \frac{22}{7} \times 7 \times 7 \times 10 = 22 \times 70 = \mathbf{1,540}\ \text{cm}^3$.
- $\text{TSA} = 2\pi r(r+h) = 2 \times \frac{22}{7} \times 7 \times (7+10) = 44 \times 17 = \mathbf{748}\ \text{cm}^2$.

MODULE 5: TRIGONOMETRY (10 Marks)

Chapter 23: Trigonometric Ratios & Heights and Distances

- **Key Trigonometric Ratios Table**: Memorize $30^\circ, 45^\circ, 60^\circ$ for $\sin, \cos, \tan$.
- **Heights & Distances**: Master the Lighthouse/Tower template (refer to Template 1 in `04_GUARANTEED_SURE_SHOT_LONG_QUESTIONS.md`).
- **Standard Solved Question**:
- *Evaluate:* $2\tan^2 45^\circ + \cos^2 30^\circ - \sin^2 60^\circ$:
  - $\tan 45^\circ = 1 \implies \tan^2 45^\circ = 1^2 = 1$
  - $\cos 30^\circ = \frac{\sqrt{3}}{2} \implies \cos^2 30^\circ = \frac{3}{4}$
  - $\sin 60^\circ = \frac{\sqrt{3}}{2} \implies \sin^2 60^\circ = \frac{3}{4}$
  - Value $= 2(1) + \frac{3}{4} - \frac{3}{4} = \mathbf{2}$.

MODULE 6: STATISTICS & PROBABILITY (12 Marks)

Chapter 25: Measure of Central Tendency

- **Direct Mean Formula**: $\bar{x} = \frac{\sum f_i x_i}{\sum f_i}$
- **Grouped Median Formula**: $\text{Median} = L + \left(\frac{\frac{N}{2} - cf}{f}\right) \times h$
- **Empirical Mode**: $\text{Mode} = 3\,\text{Median} - 2\,\text{Mean}$

Chapter 26: Probability

- **Core Formula**: $P(E) = \frac{\text{favourable outcomes}}{\text{total outcomes}}$, $0 \le P(E) \le 1$.
- **Standard Solved Questions**:
- *A single die is thrown. Find the probability of getting (i) a prime number, (ii) an odd number*:
  - Sample space $= \{1,2,3,4,5,6\}$ (Total = 6).
  - (i) Prime numbers $= \{2,3,5\}$ (3 outcomes) $\implies P = \frac{3}{6} = \mathbf{\frac{1}{2}}$.
  - (ii) Odd numbers $= \{1,3,5\}$ (3 outcomes) $\implies P = \frac{3}{6} = \mathbf{\frac{1}{2}}$.

- *One card is drawn from a well-shuffled deck of 52 cards. Find the probability of getting a red queen*:
    - Red Queens $= 2$ (Queen of hearts, Queen of diamonds).
    - $P = \frac{2}{52} = \mathbf{\frac{1}{26}}$.